Mod-4: Electrodynamics

(Vector Operators, Integral Theorems, and Derivations of Maxwell's Equations)

Part I: Core Vector Differential Operator and its Operations

The Del (Nabla) operator in 3D Cartesian coordinates is defined as:

$$\nabla \equiv \hat{\mathbf{i}}\frac{\partial}{\partial x} + \hat{\mathbf{j}}\frac{\partial}{\partial y} + \hat{\mathbf{k}}\frac{\partial}{\partial z}$$

1. Gradient (∇f)

- **Mathematical Definition:** Acts on a scalar field $f(x, y, z)$ to produce a vector field:

$$\nabla f = \frac{\partial f}{\partial x}\hat{\mathbf{i}} + \frac{\partial f}{\partial y}\hat{\mathbf{j}} + \frac{\partial f}{\partial z}\hat{\mathbf{k}}$$

- **Physical Significance:** Points in the direction of the greatest spatial rate of increase of the scalar quantity. Its magnitude gives the maximum rate of spatial change (slope) per unit distance.
- **Physical Example:** *Electrostatic Potential & Field.* The electric field is the negative gradient of electric potential, $\mathbf{E} = -\nabla V$. Positive charges naturally move down the potential hill (opposite to the gradient of potential), from higher to lower potential.

2. Divergence ($\nabla \cdot \mathbf{A}$)

- **Mathematical Definition:** Scalar dot product of ∇ with a vector field $\mathbf{A}(x, y, z)$:

$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

- **Physical Significance:** Measures the net volumetric flux per unit volume leaving an infinitesimal region around a point.
  - $\nabla \cdot \mathbf{A} > 0$: Point acts as a **source** (field lines diverge outward).
  - $\nabla \cdot \mathbf{A} < 0$: Point acts as a **sink** (field lines converge inward).
  - $\nabla \cdot \mathbf{A} = 0$: **Solenoidal** field (what enters must leave; no net creation/annihilation).
- **Physical Example:** *Fluid Dynamics / Incompressible Flow.* For fluid flow with velocity $\mathbf{v}$, $\nabla \cdot \mathbf{v} > 0$ marks a water faucet expanding outwards, while $\nabla \cdot \mathbf{B} = 0$ signifies magnetic fields have no net poles (monopoles).

3. Curl ($\nabla \times \mathbf{A}$)

- **Mathematical Definition:** Vector cross product of ∇ with a vector field $\mathbf{A}$:

$$\nabla \times \mathbf{A} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)\hat{\mathbf{i}} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)\hat{\mathbf{j}} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\hat{\mathbf{k}}$$

- **Physical Significance:** Measures the microscopic circulation (vorticity/rotation) of a vector field per unit area around an infinitesimal loop. If an infinitesimal paddle wheel placed in the vector field tends to spin, the field possesses non-zero curl along the wheel's axis. Fields with $\nabla \times \mathbf{A} = \mathbf{0}$ are **irrotational** or **conservative**.

- **Physical Example:** *Whirlpools & Magnetic Circulation.* A whirlpool exhibits high curl at its center. In magnetostatics, an electric current density $\mathbf{J}$ generates a curling magnetic field circling around the wire: $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$.

Part II: Foundational Vector Integral Theorems

The transformations between the differential and integral formulations rely on two integral theorems:

1. **Divergence Theorem (Gauss's Theorem):** Converts a boundary surface integral over a closed surface V into a volume integral throughout the interior volume V:
$$\oint_S \mathbf{A} \cdot d\mathbf{S} = \int_V (\nabla \cdot \mathbf{A})\, dV$$
2. **Stokes' Theorem (Kelvin–Stokes Theorem):** Converts a closed line circulation integral around boundary path S into an open surface flux integral over surface S:
$$\oint_L \mathbf{A} \cdot d\boldsymbol{\ell} = \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S}$$

Part III: Derivation of Maxwell's Four Equations

1. Gauss's Law for Electricity

Physical Meaning: Electric flux passing through any closed surface is proportional to the total net electric charge enclosed by that surface.

Integral Form Derivation: From Coulomb's Law, the electric field $\mathbf{E}(\mathbf{r})$ of a point source charge q located at the origin is:
$$\mathbf{E}(\mathbf{r}) = \frac{q}{4\pi\varepsilon_0} \frac{\hat{\mathbf{r}}}{r^2}$$
Evaluating net flux through a closed surface S surrounding charge q:
$$\Phi_E = \oint_S \mathbf{E} \cdot d\mathbf{S} = \oint_S \frac{q}{4\pi\varepsilon_0 r^2} \hat{\mathbf{r}} \cdot (\hat{\mathbf{n}}\, dS)$$
Using differential solid angle $d\Omega = \frac{\hat{\mathbf{r}} \cdot \hat{\mathbf{n}}\, dS}{r^2}$ and closed solid angle $\oint d\Omega = 4\pi$:
$$\oint_S \mathbf{E} \cdot d\mathbf{S} = \frac{q}{4\pi\varepsilon_0} \oint d\Omega = \frac{q}{\varepsilon_0}$$
For a continuous volume charge density $\rho(\mathbf{r})$, total enclosed charge is $Q_{\text{enc}} = \int_V \rho\, dV$. By superposition:
$$\oint_{\mathbf{S}} \mathbf{E} \cdot \mathbf{dS} = \frac{\mathbf{1}}{\boldsymbol{\varepsilon_0}} \int_{\mathbf{V}} \rho\, \mathbf{dV}$$

Differential Form Derivation: Applying the Divergence Theorem to the left-hand side:
$$\int_V (\nabla \cdot \mathbf{E})\, dV = \frac{1}{\varepsilon_0} \int_V \rho\, dV \implies \int_V \left(\nabla \cdot \mathbf{E} - \frac{\rho}{\varepsilon_0} \right) dV = 0$$
Since this must hold for any arbitrarily chosen control volume V, the integrand must vanish pointwise:
$$\nabla \cdot \mathbf{E} = \frac{\rho}{\boldsymbol{\varepsilon_0}}$$

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2. Gauss's Law for Magnetism

Physical Meaning: Isolated magnetic monopoles do not exist; magnetic field lines are continuous solenoidal closed loops.

Integral Form Derivation: Magnetic fields arise from current loops or intrinsic dipoles. Because field lines have neither source nor sink, any flux entering a closed volume must also emerge:

$$\oint_{\mathbf{S}} \mathbf{B} \cdot \mathbf{dS} = \mathbf{0}$$

Differential Form Derivation: Transform the closed surface integral using the Divergence Theorem:

$$\oint_S \mathbf{B} \cdot d\mathbf{S} = \int_V (\nabla \cdot \mathbf{B})\, dV = 0 \implies \nabla \cdot \mathbf{B} = \mathbf{0}$$

Consequence (NB): Because $\nabla \cdot (\nabla \times \mathbf{A}) \equiv 0$, this implies $\mathbf{B}$ can always be written as $\mathbf{B} = \nabla \times \mathbf{A}$ (magnetic vector potential).

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3. Faraday's Law of Induction

Physical Meaning: A time-varying magnetic field induces an electric field with non-zero circulation (electromotive force).

Integral Form Derivation: Experimentally, induced EMF $\mathcal{E}$ along loop $C = L$ bounding surface S opposes the rate of change of magnetic flux Φ_B (Faraday–Lenz law):

$$\mathcal{E} = -\frac{d\Phi_B}{dt}, \quad \text{where } \mathcal{E} = \oint_L \mathbf{E} \cdot d\boldsymbol{\ell} \quad \text{and} \quad \Phi_B = \int_S \mathbf{B} \cdot d\mathbf{S}$$

For a stationary loop, bringing the time derivative inside gives:

$$\oint_{\mathbf{L}} \mathbf{E} \cdot \mathbf{d}\boldsymbol{\ell} = -\int_{\mathbf{S}} \frac{\partial \mathbf{B}}{\partial \mathbf{t}} \cdot \mathbf{dS}$$

Differential Form Derivation: Convert the line integral via Stokes' Theorem:

$$\oint_L \mathbf{E} \cdot d\boldsymbol{\ell} = \int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{S} \implies \int_S \left(\nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} \right) \cdot d\mathbf{S} = 0$$

Since surface S is arbitrary:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial \mathbf{t}}$$

4. Ampère–Maxwell Law

Physical Meaning: Magnetic fields are generated both by conduction current density $\mathbf{J}$ and by time-varying electric displacement current density.

Limitation of Ampère's Static Law: Ampère's static circuital law states:

$$\oint_L \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I_{encl}$$

For a current carrying wire in a closed loop, we know that $I_{encl} = \int_S \mathbf{J} \cdot d\mathbf{S}$. By using Stoke's theorem we can write,

$$\oint_L \mathbf{B} \cdot d\boldsymbol{\ell} = \int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{S}$$

$$\therefore \qquad \int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{S} = \mu_0 \int_S \mathbf{J} \cdot d\mathbf{S} \implies \nabla \times \mathbf{B} = \mu_0 \mathbf{J}$$

$$\therefore \qquad \int_S [(\nabla \times \mathbf{B}) - \mu_0 \mathbf{J}] \cdot d\mathbf{S} = 0$$

Since this must hold for any arbitrarily chosen control surface S, the integrand must vanish pointwise:

$$\implies \nabla \times \mathbf{B} = \mu_0 \mathbf{J}$$

Continuity Equation

Taking the divergence of both sides:

$$\nabla \cdot (\nabla \times \mathbf{B}) = \mu_0 (\nabla \cdot \mathbf{J})$$

Because the divergence of any curl is identically zero ($\nabla \cdot (\nabla \times \mathbf{V}) = 0$), this implies $\nabla \cdot \mathbf{J} = 0$. However, conservation of charge requires the **Continuity Equation**:

$$\nabla \cdot \mathbf{J} + \frac{\partial \rho}{\partial t} = 0 \implies \nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t}$$

This creates a contradiction whenever charge density varies in time ($\partial \rho / \partial t \neq 0$).

Displacement Current & Differential Form: Substitute Gauss's law $\rho = \varepsilon_0 (\nabla \cdot \mathbf{E})$ into the continuity equation:

$$\nabla \cdot \mathbf{J} = -\frac{\partial}{\partial t} (\varepsilon_0 \nabla \cdot \mathbf{E}) = -\nabla \cdot \left(\varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \implies \nabla \cdot \left(\mathbf{J} + \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) = 0$$

Defining the **displacement current density** $\mathbf{J}_D = \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}$, Maxwell corrected Ampère's law:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial \mathbf{t}}$$

Integral Form Derivation: Integrate over surface S bounded by loop L and apply Stokes' Theorem:

$$\int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{S} = \mu_0 \int_S \mathbf{J} \cdot d\mathbf{S} + \mu_0 \varepsilon_0 \int_S \frac{\partial \mathbf{E}}{\partial t} \cdot d\mathbf{S}$$

$$\oint_{\mathbf{L}} \mathbf{B} \cdot \mathbf{d}\boldsymbol{\ell} = \mu_0 \mathbf{I}_{\text{enc}} + \mu_0 \varepsilon_0 \frac{\mathbf{d}}{\mathbf{dt}} \int_{\mathbf{S}} \mathbf{E} \cdot \mathbf{dS}$$

Part IV: Summary Reference Table

Law	Differential Form	Integral Form
Gauss's Law (Electrostatics)	$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$	$\oint_S \mathbf{E} \cdot d\mathbf{S} = \frac{Q_{\text{enc}}}{\varepsilon_0}$
Gauss's Law (Magnetostatics)	$\nabla \cdot \mathbf{B} = 0$	$\oint_S \mathbf{B} \cdot d\mathbf{S} = 0$
Faraday's Law	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	$\oint_L \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d}{dt}\int_S \mathbf{B} \cdot d\mathbf{S}$
Ampère–Maxwell Law	$\nabla \times \mathbf{B} = \mu_0\mathbf{J} + \mu_0\varepsilon_0\frac{\partial \mathbf{E}}{\partial t}$	$\oint_L \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I_{\text{enc}} + \mu_0\varepsilon_0\frac{d}{dt}\int_S \mathbf{E} \cdot d\mathbf{S}$

Part V: Comprehensive Free Space vs. Medium Comparison Table

Law	Microscopic Form (Free Space / Vacuum)	Macroscopic Form (In a Medium)
Gauss's Law ($\mathbf{E}, \mathbf{D}$)	$\nabla \cdot \mathbf{E} = \frac{\rho_{\text{f}}}{\varepsilon_0}$	$\nabla \cdot \mathbf{D} = \rho_{total}$
Gauss's Law ($\mathbf{B}$)	$\nabla \cdot \mathbf{B} = 0$	$\nabla \cdot \mathbf{B} = 0$
Faraday's Law	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$
Ampère–Maxwell Law	$\nabla \times \mathbf{B} = \mu_0\mathbf{J}_{\text{f}} + \mu_0\varepsilon_0\frac{\partial \mathbf{E}}{\partial t}$	$\nabla \times \mathbf{H} = \mathbf{J}_{total} + \frac{\partial \mathbf{D}}{\partial t}$
Constitutive Links	$\mathbf{D} = \varepsilon_0\mathbf{E}, \quad \mathbf{B} = \mu_0\mathbf{H}$	$\mathbf{D} = \varepsilon_0\mathbf{E} + \mathbf{P}, \quad \mathbf{H} = \frac{1}{\mu_0}\mathbf{B} - \mathbf{M}$

The total electric charge density ρ_{total} consists of:

$$\rho_{\text{total}} = \rho_f + \rho_b$$

$$\mathbf{J}_{\text{total}} = \mathbf{J}_f + \mathbf{J}_b + \mathbf{J}_p$$

To close the system of macroscopic equations, material-specific **constitutive relations** are introduced:

$$\mathbf{P} = \varepsilon_0\chi_e\mathbf{E} \quad \Longrightarrow \quad \mathbf{D} = \varepsilon_0(1+\chi_e)\mathbf{E} = \varepsilon_0\varepsilon_r\mathbf{E} = \varepsilon\mathbf{E}$$

$$\mathbf{M} = \chi_m\mathbf{H} \quad \Longrightarrow \quad \mathbf{B} = \mu_0(1+\chi_m)\mathbf{H} = \mu_0\mu_r\mathbf{H} = \mu\mathbf{H}$$

where χ_e and χ_m are electric and magnetic susceptibilities, $\varepsilon = \varepsilon_0\varepsilon_r$ is permittivity, and $\mu = \mu_0\mu_r$ is permeability. Additionally, for conducting media with conductivity σ, Ohm's law provides $\mathbf{J}_f = \sigma\mathbf{E}$.

Derivation's and details are out of the scope of university syllabus. Reading suggested only for deeper understanding and completeness to appreciate beauty of Maxwell's equations.

Part V: Transformation from Free Space to Material Media ($\mathbf{E} \to \mathbf{D}$, $\mathbf{B} \to \mathbf{H}$)

In vacuum (microscopic picture), all charges ρ and currents $\mathbf{J}$ are treated explicitly on the atomic level. In material media (macroscopic picture), matter responds to applied electromagnetic fields by polarizing and magnetizing, which produces internal atomic-scale bound charges and currents that are difficult to track individually. We separate total sources into **free** and **bound** components.

1. Charge Decomposition and Electric Displacement Field (D)

The total electric charge density ρ_{total} consists of:

$$\rho_{\text{total}} = \rho_f + \rho_b$$

where ρ_f is the controllable free charge density (e.g., conduction electrons), and ρ_b is the bound (polarization) charge density arising from microscopic dipole alignment:

$$\rho_b = -\nabla \cdot \mathbf{P}$$

Here, $\mathbf{P}$ is the **polarization vector** (electric dipole moment per unit volume).

Substituting ρ_{total} into the microscopic Gauss's Law:

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \frac{\rho_f + \rho_b}{\varepsilon_0} = \frac{\rho_f - \nabla \cdot \mathbf{P}}{\varepsilon_0} \\ \varepsilon_0(\nabla \cdot \mathbf{E}) + \nabla \cdot \mathbf{P} &= \rho_f \\ \nabla \cdot (\varepsilon_0 \mathbf{E} + \mathbf{P}) &= \rho_f \end{aligned}$$

Defining the **Electric Displacement Field D**:

$$\mathbf{D} \equiv \varepsilon_0 \mathbf{E} + \mathbf{P} \quad \Longrightarrow \quad \nabla \cdot \mathbf{D} = \rho_{\mathbf{f}}$$

Using the Divergence Theorem, the corresponding integral form over closed surface S is:

$$\oint_{\mathbf{S}} \mathbf{D} \cdot \mathbf{dS} = \mathbf{Q}_{\mathbf{f},\text{enc}} = \int_{\mathbf{V}} \rho_{\mathbf{f}} \, \mathbf{dV}$$

Significance: The flux of $\mathbf{D}$ depends solely on the free charges placed in the medium, absorbing internal polarization effects directly into the definition of $\mathbf{D}$.

2. Current Decomposition and Magnetic Field Intensity (H)

The total current density $\mathbf{J}_{\text{total}}$ in matter is composed of:

$$\mathbf{J}_{\text{total}} = \mathbf{J}_f + \mathbf{J}_b + \mathbf{J}_p$$

- $\mathbf{J}_f$: Free current density (drift of free charges).

- $\mathbf{J}_b = \nabla \times \mathbf{M}$: Bound (magnetization) current density due to microscopic atomic orbital and spin magnetic dipoles, where $\mathbf{M}$ is the **magnetization vector** (magnetic dipole moment per unit volume).
- $\mathbf{J}_p = \frac{\partial \mathbf{P}}{\partial t}$: Polarization current density resulting from time-dependent displacement of bound charges ($\nabla \cdot \mathbf{J}_p = -\frac{\partial \rho_b}{\partial t}$).

Substitute $\mathbf{J}_{\text{total}}$ into the microscopic Ampère–Maxwell Law:

$$\nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J}_f + \nabla \times \mathbf{M} + \frac{\partial \mathbf{P}}{\partial t} + \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right)$$

$$\frac{1}{\mu_0}(\nabla \times \mathbf{B}) - (\nabla \times \mathbf{M}) = \mathbf{J}_f + \frac{\partial}{\partial t}(\varepsilon_0 \mathbf{E} + \mathbf{P})$$

$$\nabla \times \left(\frac{1}{\mu_0} \mathbf{B} - \mathbf{M} \right) = \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t}$$

Defining the auxiliary **Magnetic Field Intensity H**:

$$\mathbf{H} \equiv \frac{1}{\mu_0} \mathbf{B} - \mathbf{M} \quad \Longrightarrow \quad \nabla \times \mathbf{H} = \mathbf{J_f} + \frac{\partial \mathbf{D}}{\partial \mathbf{t}}$$

Applying Stokes' Theorem gives the integral form:

$$\oint_{\mathbf{L}} \mathbf{H} \cdot \mathbf{d}\boldsymbol{\ell} = \mathbf{I}_{\mathbf{f},\text{enc}} + \int_{\mathbf{S}} \frac{\partial \mathbf{D}}{\partial \mathbf{t}} \cdot \mathbf{dS}$$

3. Invariance of Homogeneous Equations

Faraday's Law and Gauss's Law for Magnetism do not involve sources (ρ or $\mathbf{J}$); they describe fundamental topological and kinematic relationships between fields and space-time. Consequently, their mathematical forms remain identical in media:

$$\nabla \cdot \mathbf{B} = 0, \qquad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

4. Constitutive Relations for Linear, Isotropic, Homogeneous (LIH) Media

To close the system of macroscopic equations, material-specific **constitutive relations** are introduced:

$$\mathbf{P} = \varepsilon_0 \chi_e \mathbf{E} \quad \Longrightarrow \quad \mathbf{D} = \varepsilon_0 (1 + \chi_e) \mathbf{E} = \varepsilon_0 \varepsilon_r \mathbf{E} = \varepsilon \mathbf{E}$$

$$\mathbf{M} = \chi_m \mathbf{H} \quad \Longrightarrow \quad \mathbf{B} = \mu_0 (1 + \chi_m) \mathbf{H} = \mu_0 \mu_r \mathbf{H} = \mu \mathbf{H}$$

where χ_e and χ_m are electric and magnetic susceptibilities, $\varepsilon = \varepsilon_0 \varepsilon_r$ is permittivity, and $\mu = \mu_0 \mu_r$ is permeability. Additionally, for conducting media with conductivity σ, Ohm's law provides $\mathbf{J}_f = \sigma \mathbf{E}$.
